

Finding a Percent of a Quantity

Hundredths grids, ten-percent tape strips, and decimal multipliers for finding a percent of a quantity

Directions:

- Every problem here asks the same question in a different costume: *how much is this percent of this amount?*
- Each problem names a model — a hundredths grid, a ten-percent tape strip, or the decimal form of the percent. Show that model in your work space, then write the number it gives you on the answer line.
- Percent means “per one hundred.” Keep that sentence in view.

1. A hundredths grid is one square cut into 100 equal parts, so each part is one percent of the whole. Ms. Ruiz shades 30 parts of a grid to stand for the tomato plants in her garden of 200 plants. How many plants do the shaded parts stand for?

Answer: _____

2. On another hundredths grid, 45 of the 100 parts are shaded to stand for the novels on a shelf of 60 books. How many books do the shaded parts stand for?

Answer: _____

3. A tape diagram splits the 350 seats of a theatre into 10 equal strips, so one strip is ten percent of the seats. Fill in the tape table, then record the two amounts.

Percent of the seats	10%	30%
Number of seats	(a)	(b)

(a) seats in one ten-percent strip: _____

(b) seats in three of those strips: _____

4. A percent becomes a decimal when you divide it by 100. A rack holds 84 jackets, and 25% of them are blue.

(a) 25% written as a decimal: _____

(b) blue jackets on the rack: _____

5. A pizza night cost 60 dollars. Build the tip out of ten-percent strips instead of multiplying: draw the strip for 10%, halve it for 5%, then put the pieces together.

(a) 10% of 60 dollars: _____

(b) 5% of 60 dollars: _____

(c) 15% of 60 dollars: _____

6. A school has 250 students, and 8% of them were absent today. Write the percent as a decimal, multiply, and give the number of absent students.

Answer: _____

7. A hundredths grid stands for a 400 gram bag of trail mix, so the 100 equal parts share the bag evenly. The 35 shaded parts are the raisins.

(a) grams of trail mix in one part of the grid: _____

(b) grams of raisins in the bag: _____

8. A hiking club has 45 members. Use ten-percent strips of the membership, not a calculator.

(a) 10% of the members: _____

(b) 60% of the members: _____

9. A repair bill of 320 dollars carries a 7.5% service charge. Use the decimal form of the percent.

- (a) 7.5% written as a decimal: _____
- (b) the service charge, in dollars: _____

10. Maya says that 20% of 150 dollars must be more than 25% of 100 dollars, because 150 dollars is the larger amount to start with.

- (a) 20% of 150 dollars: _____
- (b) 25% of 100 dollars: _____
- (c) Use your two amounts to explain whether Maya's reasoning is correct. Write your answer in the work space.